Data Visualization Lab Assignment - 1

Fatima Sohail

2026-01-31

Descriptive & Predictive Analysis

Dataset Description

This dataset contains information about customers' shopping habits at Walmart. It shows what products people buy, how much they spend, and some basic details about the customers, such as their age and gender. The data can help us understand customers' preferences and shopping patterns.

It includes details like:

Customer information: IDs and demographics like age and gender.

Purchase information: Products bought, purchase amounts, and transaction records.

Purpose: The dataset can be used to analyze customer behavior, group similar customers, predict shopping trends, and visualize sales patterns.

The dataset has a large number of records, giving a clear view of Walmart's customer buying behavior. It is suitable for tasks like descriptive analysis, trend analysis, and predictive modeling.

1. Understanding the Dataset

```
# Load required libraries
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.1
```

```
## Warning: package 'ggplot2' was built under R version 4.5.1
```

```
## Warning: package 'tibble' was built under R version 4.5.1
```

```
## Warning: package 'tidyr' was built under R version 4.5.1
```

```
## Warning: package 'readr' was built under R version 4.5.1
```

```
## Warning: package 'purrr' was built under R version 4.5.1
```

```
## Warning: package 'dplyr' was built under R version 4.5.1
```

```
## Warning: package 'stringr' was built under R version 4.5.1
```

```
## Warning: package 'forcats' was built under R version 4.5.1
```

```
## Warning: package 'lubridate' was built under R version 4.5.1
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✓ dplyr     1.1.4     ✓ readr     2.1.5
## ✓ forcats   1.0.0     ✓ stringr   1.5.1
## ✓ ggplot2   3.5.2     ✓ tibble    3.3.0
## ✓ lubridate 1.9.4     ✓ tidyr     1.3.1
## ✓ purrr     1.1.0
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
e errors
```

```
library(ggplot2)
library(lubridate)


# Load dataset
walmart <- read.csv("Walmart_customer_purchases.csv")


# View structure and first few rows
str(walmart)
```

```
## 'data.frame':    50000 obs. of  12 variables:
##  $ Customer_ID     : chr  "84607c1f-910c-44d5-b89f-e1ee06dd34c0" "f2a81712-a73e-4424-8b39-4c6
15a0bd4ea" "da9be287-8b0e-4688-bccd-1a2cdd7567c6" "50ec6932-3ac7-492f-9e55-4b148212f302" ...
##  $ Age             : int  49 36 52 47 43 59 47 23 60 52 ...
##  $ Gender          : chr  "Female" "Other" "Male" "Female" ...
##  $ City            : chr  "New Cynthia" "Cruzport" "Jeffreytown" "Jenniferburgh" ...
##  $ Category        : chr  "Electronics" "Clothing" "Beauty" "Electronics" ...
##  $ Product_Name    : chr  "Smartphone" "T-Shirt" "Perfume" "Smartwatch" ...
##  $ Purchase_Date   : chr  "2024-08-30" "2024-12-21" "2024-12-26" "2024-11-04" ...
##  $ Purchase_Amount : num  253.3 73.2 125.6 450.3 369.3 ...
##  $ Payment_Method  : chr  "Cash on Delivery" "Debit Card" "Credit Card" "Credit Card" ...
##  $ Discount_Applied: chr  "No" "Yes" "Yes" "No" ...
##  $ Rating          : int  1 1 1 2 2 3 4 2 5 3 ...
##  $ Repeat_Customer : chr  "Yes" "No" "No" "Yes" ...
```

```
head(walmart)
```

```
##                            Customer_ID Age Gender            City    Category
## 1 84607c1f-910c-44d5-b89f-e1ee06dd34c0  49 Female     New Cynthia Electronics
## 2 f2a81712-a73e-4424-8b39-4c615a0bd4ea  36  Other        Cruzport    Clothing
## 3 da9be287-8b0e-4688-bccd-1a2cdd7567c6  52   Male     Jeffreytown      Beauty
## 4 50ec6932-3ac7-492f-9e55-4b148212f302  47 Female   Jenniferburgh Electronics
## 5 8fdc3098-fc75-4b0f-983c-d8d8168c6362  43  Other       Kingshire Electronics
## 6 fdf82943-1b5f-4e0a-bf42-9a8e2035924e  59 Female North Josemouth Electronics
##   Product_Name Purchase_Date Purchase_Amount   Payment_Method Discount_Applied
## 1   Smartphone    2024-08-30          253.26 Cash on Delivery              No
## 2      T-Shirt    2024-12-21           73.19       Debit Card             Yes
## 3      Perfume    2024-12-26          125.62      Credit Card             Yes
## 4   Smartwatch    2024-11-04          450.32      Credit Card              No
## 5   Smartphone    2024-10-07          369.28      Credit Card             Yes
## 6       Laptop    2024-02-15          435.48 Cash on Delivery              No
##   Rating Repeat_Customer
## 1      1             Yes
## 2      1              No
## 3      1              No
## 4      2             Yes
## 5      2             Yes
## 6      3              No
```

```
summary(walmart)
```

```
##  Customer_ID             Age           Gender             City
##  Length:50000       Min.   :18.00   Length:50000       Length:50000
##  Class :character   1st Qu.:28.00   Class :character   Class :character
##  Mode  :character   Median :39.00   Mode  :character   Mode  :character
##                     Mean   :38.95
##                     3rd Qu.:50.00
##                     Max.   :60.00
##    Category         Product_Name       Purchase_Date      Purchase_Amount
##  Length:50000       Length:50000       Length:50000       Min.   : 10.01
##  Class :character   Class :character   Class :character   1st Qu.:133.05
##  Mode  :character   Mode  :character   Mode  :character   Median :255.04
##                                                           Mean   :255.53
##                                                           3rd Qu.:378.91
##                                                           Max.   :499.99
##  Payment_Method     Discount_Applied       Rating      Repeat_Customer
##  Length:50000       Length:50000       Min.   :1.000   Length:50000
##  Class :character   Class :character   1st Qu.:2.000   Class :character
##  Mode  :character   Mode  :character   Median :3.000   Mode  :character
##                                        Mean   :2.999
##                                        3rd Qu.:4.000
##                                        Max.   :5.000
```

Explanation: This code loads the dataset and displays its structure, first few records, and summary statistics. It helps understand each column, data types, and ranges of values.

Observations: The dataset contains customer demographics such as age, gender, and city category, along with product categories and purchase amount. Some columns may have missing values, which are checked next.

```
# Check missing values
colSums(is.na(walmart))
```

```
##      Customer_ID              Age           Gender             City
##                0                0                0                0
##         Category     Product_Name    Purchase_Date  Purchase_Amount
##                0                0                0                0
##   Payment_Method Discount_Applied           Rating  Repeat_Customer
##                0                0                0                0
```

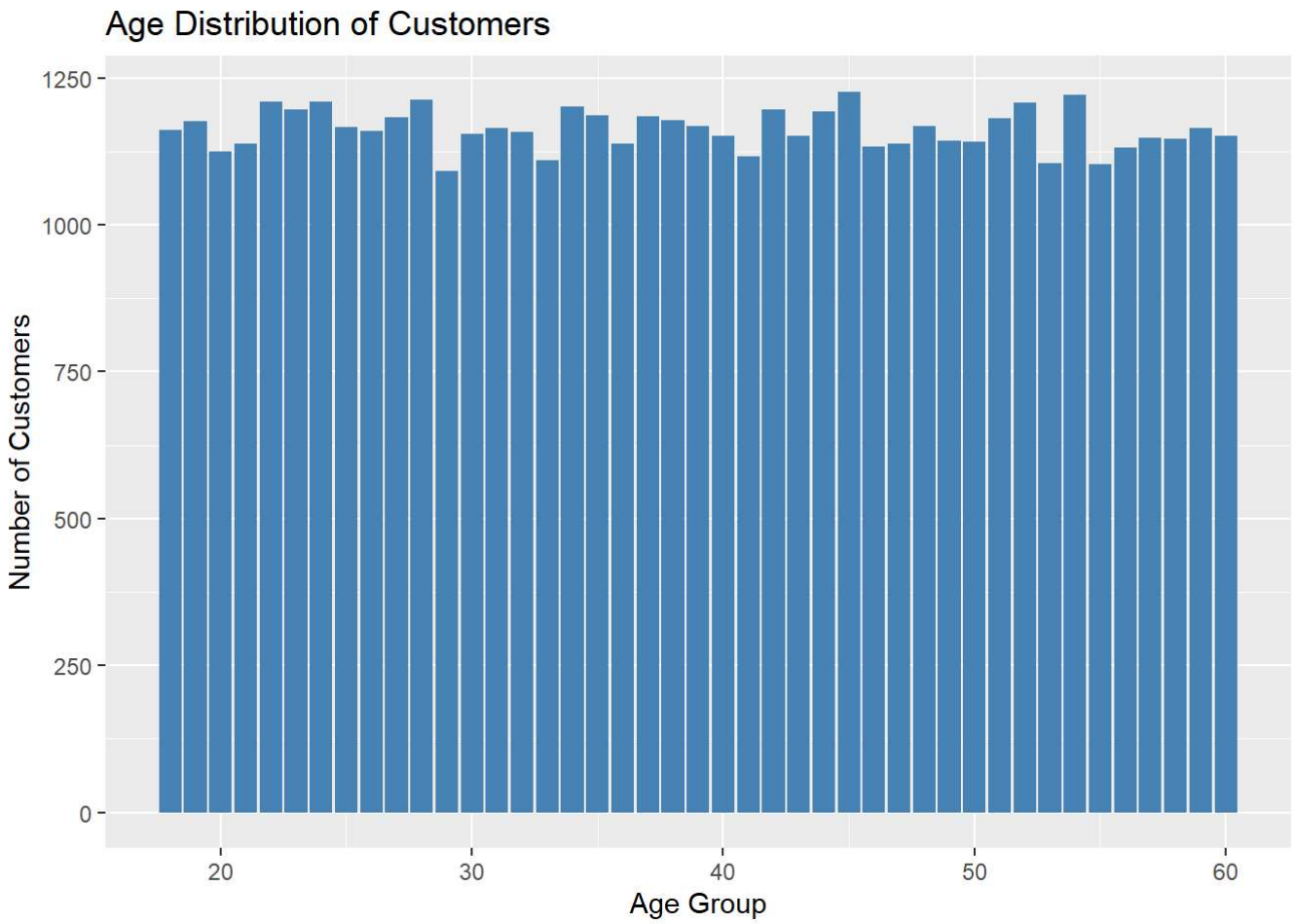
Explanation: This code checks for missing values in each column.

Insight: The columns have no missing values, making the dataset suitable for analysis without heavy preprocessing.

2. Exploring Customer Demographics

Age Distribution

```
ggplot(walmart, aes(x = Age)) +
geom_bar(fill = "steelblue") +
labs(title = "Age Distribution of Customers",
x = "Age Group",
y = "Number of Customers")
```

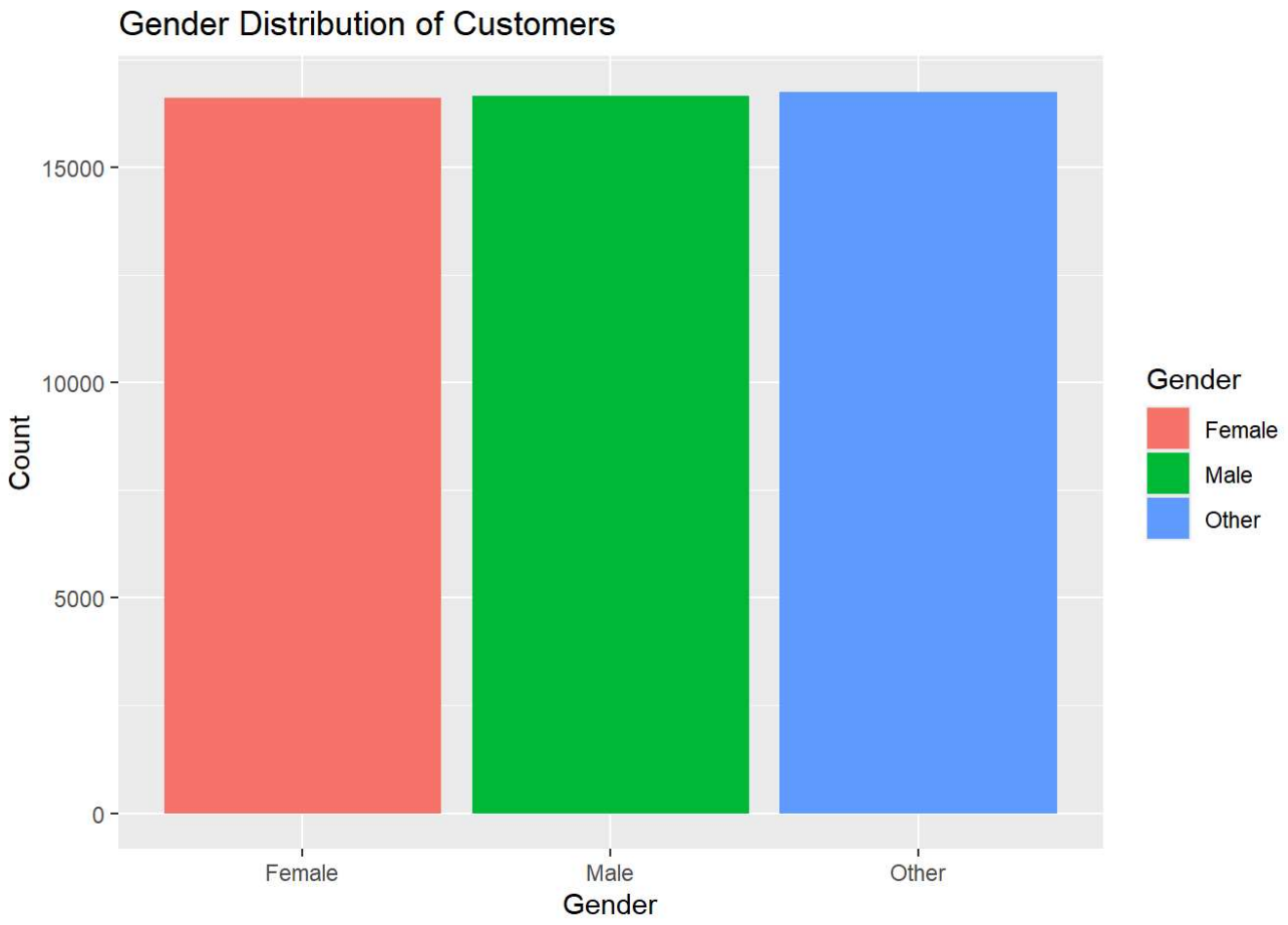


Explanation: This bar chart shows how customers are distributed across different age groups.

Insight: Middle-aged customers (26–45) appear to be the most active shoppers.

Gender Distribution

```
ggplot(walmart, aes(x = Gender, fill = Gender)) +
geom_bar() +
labs(title = "Gender Distribution of Customers",
x = "Gender",
y = "Count")
```



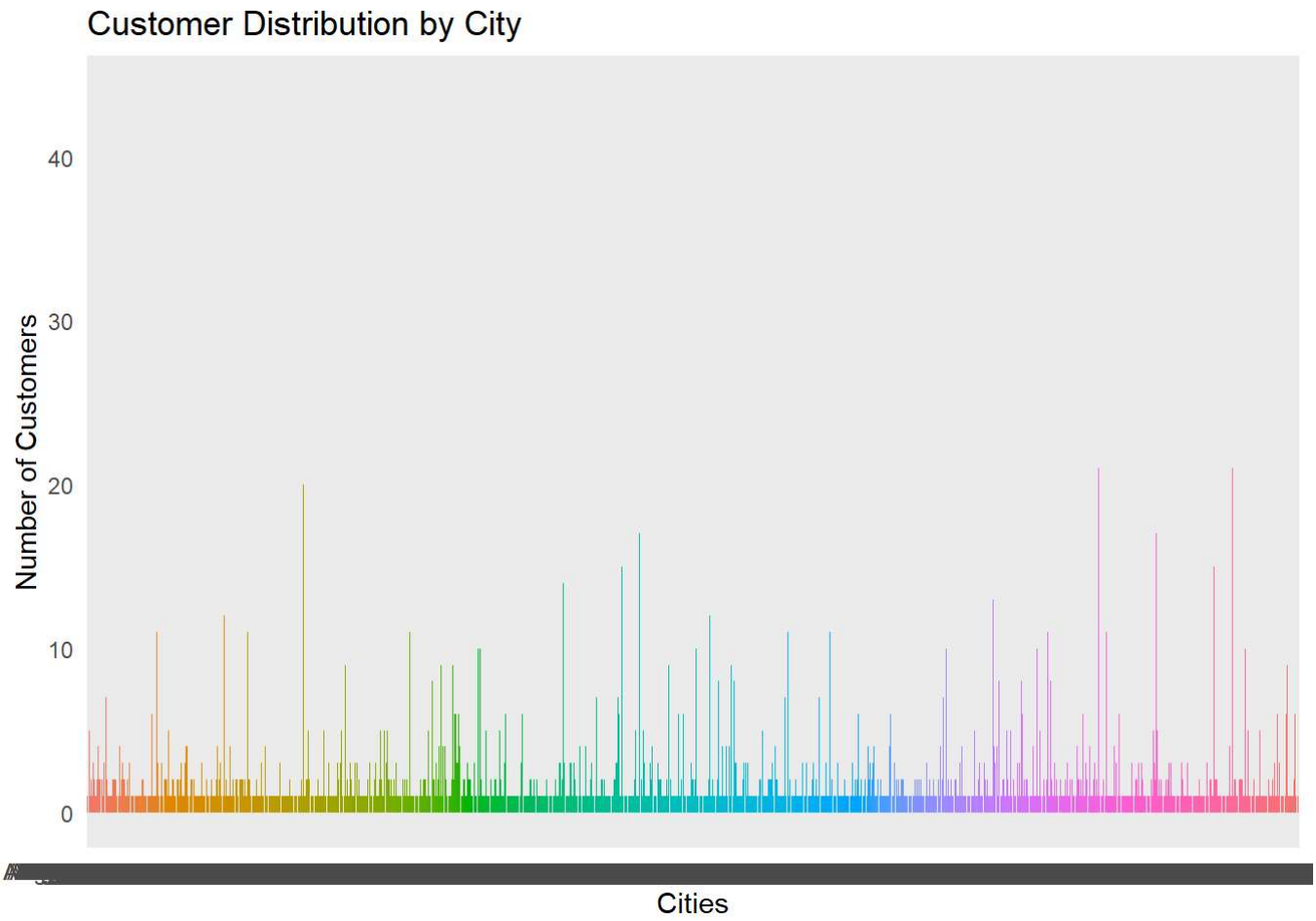
Explanation: This visualization compares the number of male and female customers.

Insight: Male customers slightly dominate the purchasing activity in this dataset

City-wise Activity

```
# Convert City to factor
walmart$City <- as.factor(walmart$City)

# Barplot
ggplot(walmart, aes(x = City, fill = City)) +
  geom_bar() +
  labs(title = "Customer Distribution by City",
       x = "Cities",
       y = "Number of Customers") +
  theme_minimal() +
  theme(legend.position = "none")  # remove legend for faster rendering
```



Explanation: This visualization compares customer activity across different cities.

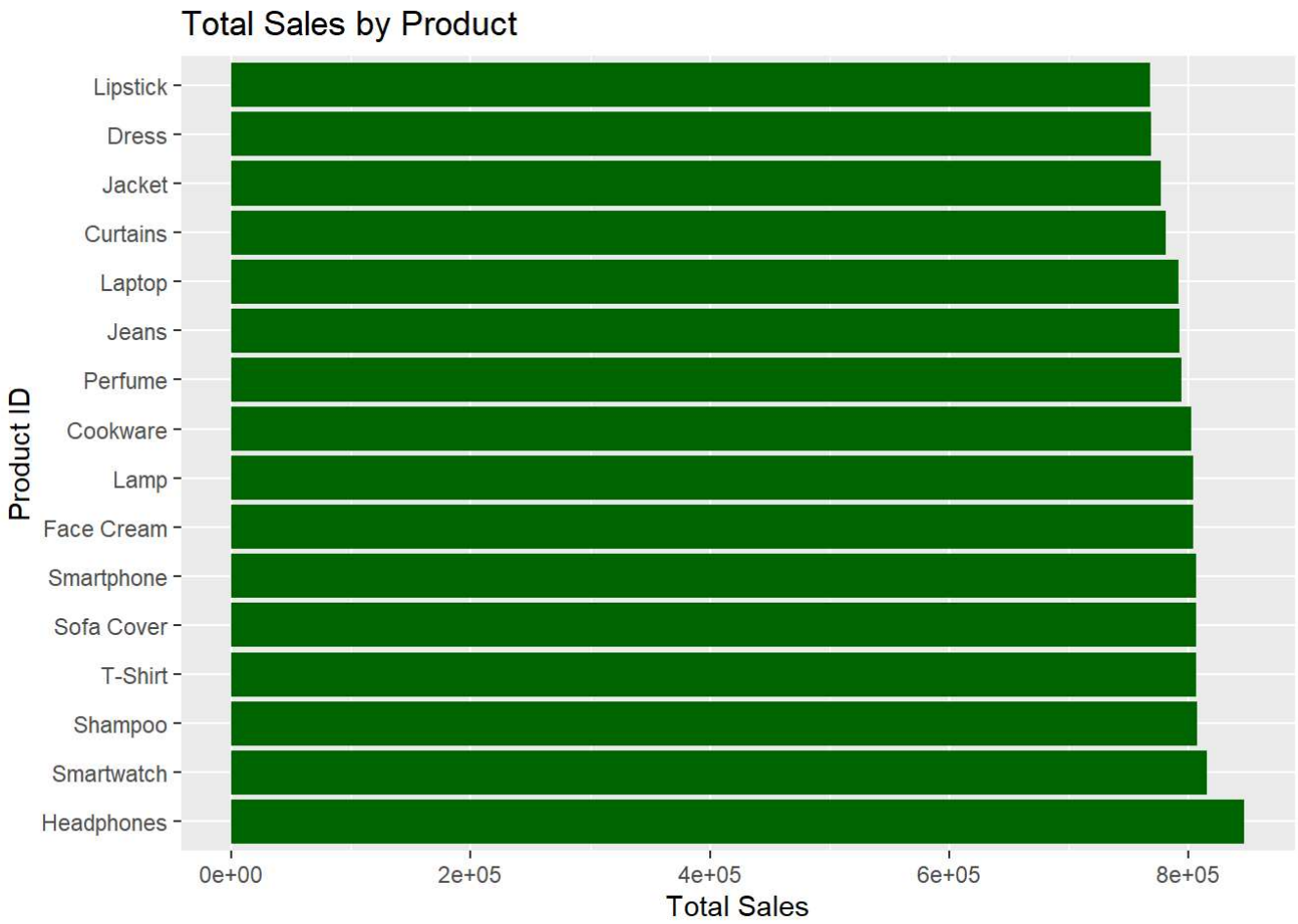
Insight: City category B has the highest customer participation.

3. Sales and Product Category Analysis

Total Sales by Product Category

```
# Total Sales by Product (Fixed)
sales_category <- walmart %>%
  group_by(Product_Name) %>%                      # Use actual product column
  summarise(Total_Sales = sum(Purchase_Amount, na.rm = TRUE))  # Use actual purchase column

# Plotting
ggplot(sales_category,
       aes(x = reorder(Product_Name, -Total_Sales),   # x-axis = Product
           y = Total_Sales)) +
  geom_col(fill = "darkgreen") +
  coord_flip() +
  labs(title = "Total Sales by Product",
       x = "Product ID",
       y = "Total Sales")
```



Explanation: This code calculates total sales for each product category and visualizes them.

Insight: Some product categories perform significantly better than others.

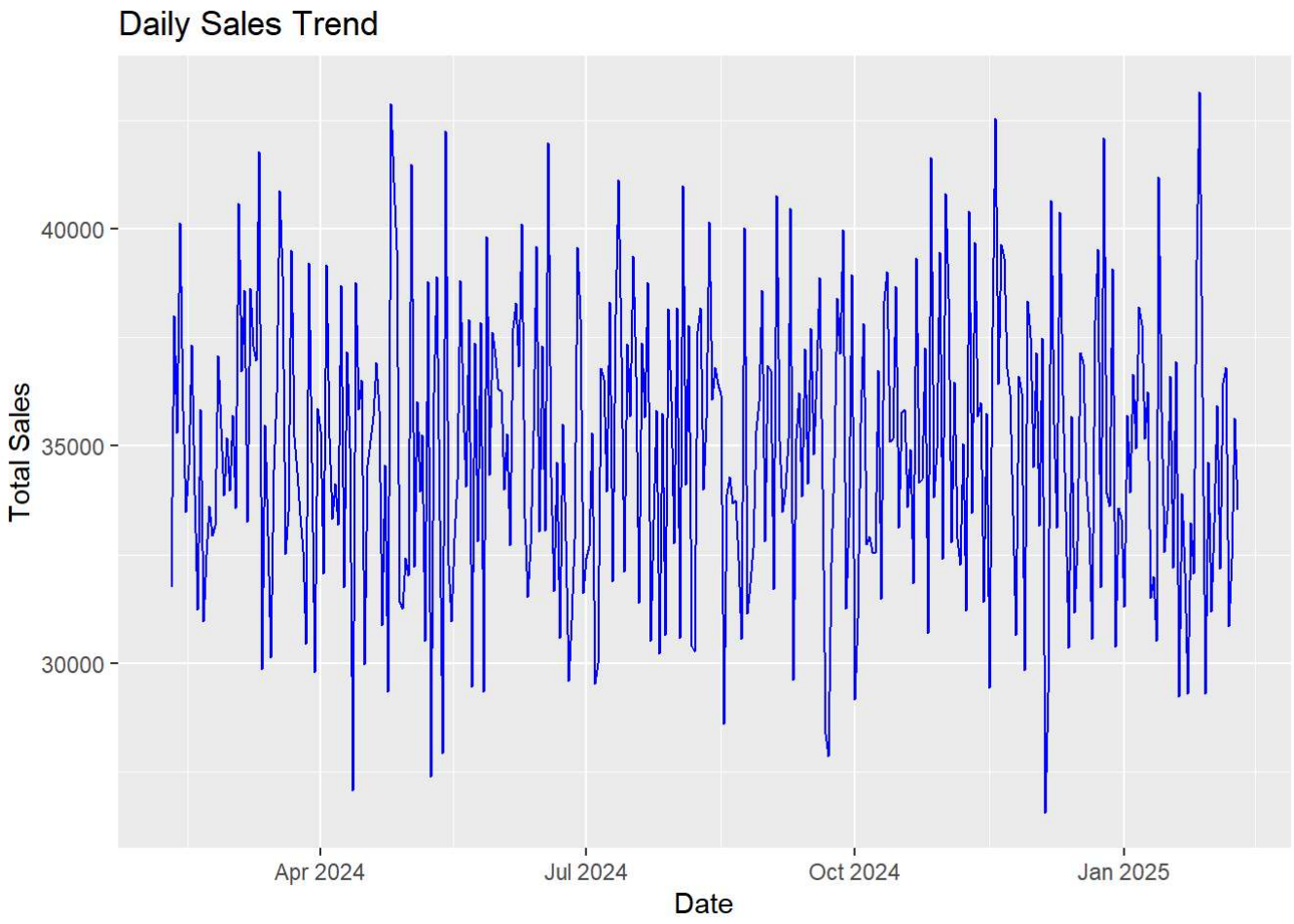
4. Time-Based Trends

Sales Over Time

```
# Convert purchase date to Date format
walmart$Purchase_Date <- as.Date(walmart$Purchase_Date)

daily_sales <- walmart %>%
  group_by(Purchase_Date) %>%
  summarise(Total_Sales = sum(Purchase_Amount))

ggplot(daily_sales, aes(x = Purchase_Date, y = Total_Sales)) +
  geom_line(color = "blue") +
  labs(title = "Daily Sales Trend",
       x = "Date",
       y = "Total Sales")
```



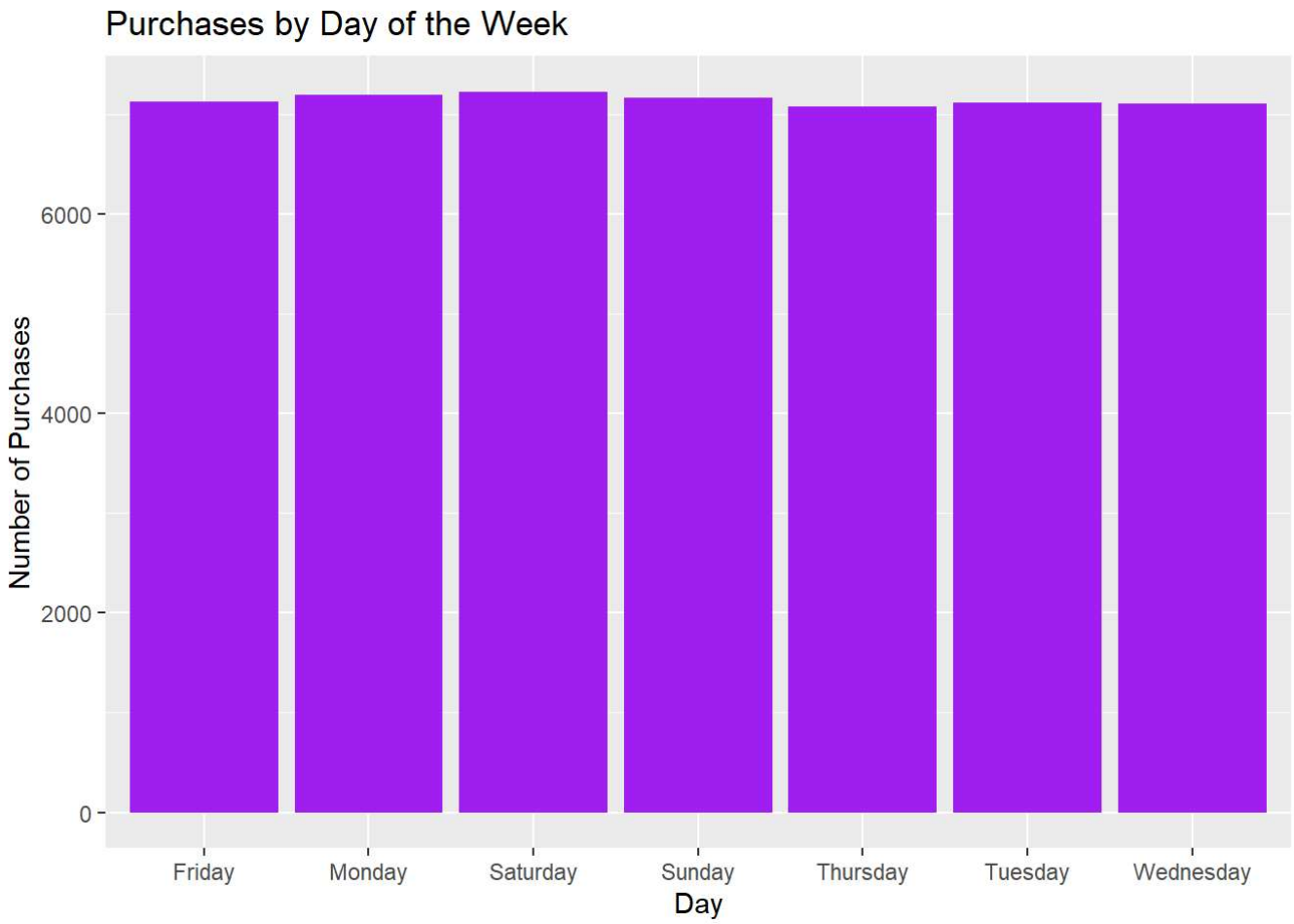
Explanation: This code analyzes how sales change over time using the purchase date.

Insight: Sales fluctuate across dates, indicating peak and low shopping periods.

Peak Shopping Days

```
walmart$Day <- weekdays(walmart$Purchase_Date)

ggplot(walmart, aes(x = Day)) +
  geom_bar(fill = "purple") +
  labs(title = "Purchases by Day of the Week",
       x = "Day",
       y = "Number of Purchases")
```



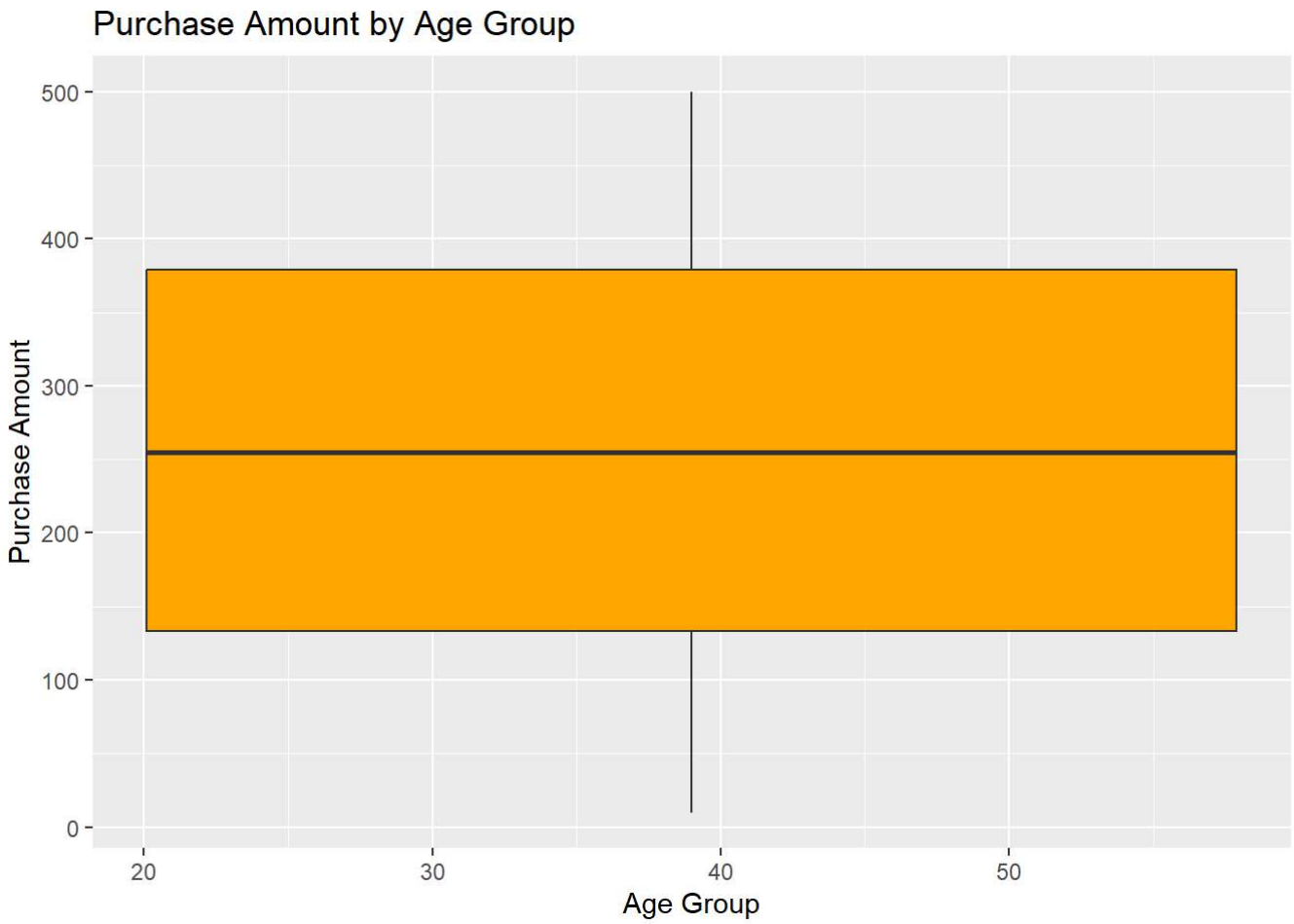
Insight: Certain days show higher purchase activity, indicating peak shopping days.

5. Advanced Insights

Purchase Amount by Age Group

```
ggplot(walmart, aes(x = Age, y = Purchase_Amount)) +
  geom_boxplot(fill = "orange") +
  labs(title = "Purchase Amount by Age Group",
       x = "Age Group",
       y = "Purchase Amount")
```

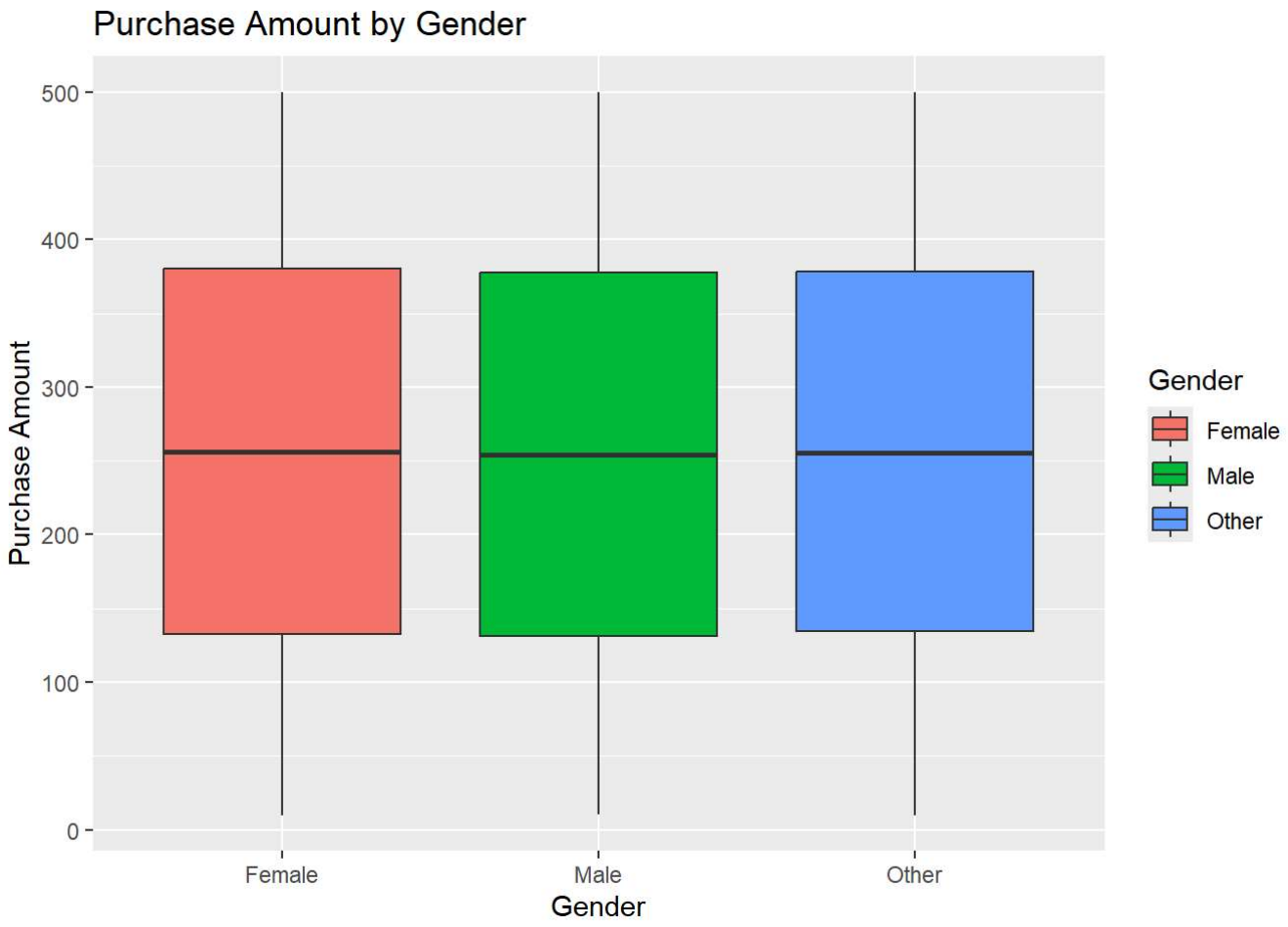
```
## Warning: Continuous x aesthetic
## ℹ did you forget `aes(group = ...)`?
```



Insight: Middle-aged customers tend to have higher median purchase values.

Purchase Behavior by Gender

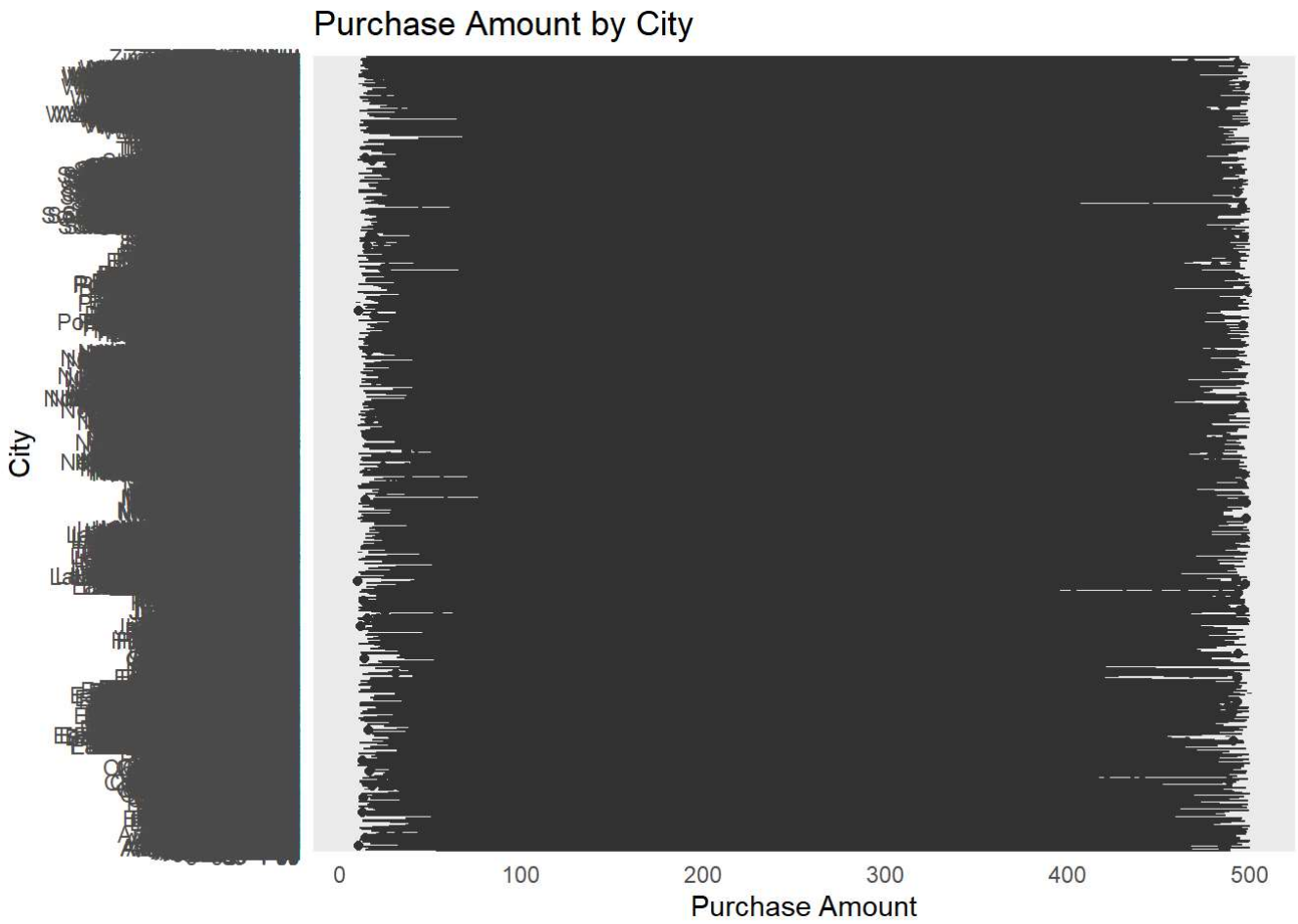
```
ggplot(walmart, aes(x = Gender, y = Purchase_Amount, fill = Gender)) +
  geom_boxplot() +
  labs(title = "Purchase Amount by Gender",
       x = "Gender",
       y = "Purchase Amount")
```



Insight: Male customers show slightly higher spending patterns.

Product Preference by City

```
# Boxplot
ggplot(walmart, aes(x = City, y = Purchase_Amount, fill = City)) +
  geom_boxplot() +
  coord_flip() +  # horizontal to improve readability
  labs(title = "Purchase Amount by City",
       x = "City",
       y = "Purchase Amount") +
  theme_minimal() +
  theme(legend.position = "none")  # optional: hide legend
```



Insight: Customers from certain cities spend more per purchase.

6. Insights and Recommendations

Key Findings

Customers aged 26–45 are the most active buyers.

Certain product categories generate the highest sales.

Sales vary over time with identifiable peak days.

Male customers have slightly higher purchase amounts.

Recommendations

Target marketing campaigns toward high-performing age groups.

Focus promotions on top-selling product categories.

Increase inventory and offers during peak shopping days.

Customize city-based promotions for high-spending locations.

7. Conclusion

This analysis highlights how customer demographics, product categories, and time-based trends influence purchasing behavior. The insights gained from visualizations can help improve sales strategies and business decision-making.